Princeton, September 25, 1987

Dear Maxim,

Inspired by a conversation with Witten, I can now compactify - with a divisor with normal crossings at infinity - the moduli stack of super Riemann surfaces. As I have to think, I continue in French.

1. Generically Smooth Cohen-Macaulay Supercurves

In this section I propose an analogue, in algebraic supergeometry, of the notion of a geometrically reduced curve.

The accepted notion of a smooth supercurve X/S is the following: a superscheme X over the superscheme S, smooth of relative dimension $(1,1)$, equipped with

$$\mathcal{D} \subset T_{X/S}$$

locally a direct factor of dimension $(0,1)$, and "as non-integrable as possible." For such a system $(X, \mathcal{D})$, one identifies $T_{X/S}/\mathcal{D}$ with $\mathcal{D}^{\otimes 2}$ by

$$D_1 \otimes D_2 \longmapsto \frac{1}{2}[D_1, D_2].$$

Dualizing $\mathcal{D} \subset T_{X/S}$ gives

$$\Omega^1_{X/S} \longrightarrow \mathcal{D}^\vee, \tag{1.1}$$

and the isomorphism $T_{X/S}/\mathcal{D} \simeq \mathcal{D}^{\otimes 2}$ dualizes to

$$0 \longrightarrow \mathcal{D}^{\vee \otimes 2} \longrightarrow \Omega^1_{X/S} \longrightarrow \mathcal{D}^\vee \longrightarrow 0,$$

whence an isomorphism

$$\mathrm{Ber}(\Omega^1_{X/S}) = \mathcal{D}^\vee.$$

Thus one obtains

$$\delta : \Omega^1_{X/S} \longrightarrow \omega_{X/S} := \mathrm{Ber}(\Omega^1_{X/S}), \tag{1.2}$$

that is, a derivation

$$\delta : \mathcal{O}_X \longrightarrow \omega_{X/S}.$$

My choice of signs may be ambiguous. For X with local coordinates (z, θ), with

$$D = \partial_\theta + \theta \partial_z,$$

I want

$$\delta : f \longmapsto \big((\partial_\theta + \theta\partial_z) f\big)[dz/d\theta],$$

where $[dz/d\theta]$ is the basis of $\mathrm{Ber}(\Omega^1_{X/S})$ defined by the basis $(dz, d\theta)$ of $\Omega^1_{X/S}$.

Conversely, δ determines $\mathcal{D}$: $\mathcal{D}$ is the orthogonal of $\mathrm{Ker}(\delta)$, or the multiples of δ in the basis of $\mathrm{Ber}(\Omega^1_{X/S})$. Notice that, for an epimorphism $\delta : \Omega^1_{X/S} \to \omega_X$, it is not enough that $\mathrm{Ker}(\delta)^\perp$ be as non-integrable as possible for δ to come from a supercurve structure: the normalization must be correct,

$$\delta f = (Df)\ell, \qquad \ell = [D^2/D]^{-1}.$$

In local coordinates (z, θ), for

$$D := \partial_\theta + \theta \partial_z$$

(an odd derivation), one has:

Lemma 1.1. For

$$\delta = (aD + bD^2)[dz/d\theta]$$

to define a supercurve structure, it is necessary and sufficient that a be invertible and that

$$[(aD + bD^2)^2/(aD + bD^2)] = [D^2/D],$$

i.e.

$$\mathrm{Ber}\begin{pmatrix} a^2 + bD^2b + aDb & b \\ -(aDa + bD^2a) & a \end{pmatrix} = 1, \tag{1.1.1}$$

where

$$\mathrm{Ber}\begin{pmatrix} a & \beta \\ \gamma & d \end{pmatrix} = (ad - \beta\gamma)d^{-2}.$$

Proof. One has

$$(aD + bD^2)^2 = (aDa + bD^2a)D + (a^2 + bD^2b + aDb)D^2.$$

The case that we shall use most is the following: $\delta = D\ell$ is a supercurve structure, $S_0 \hookrightarrow S$ is defined by an ideal of square zero, and

$$D_1 = D + a_1 D + b_1 D^2$$

coincides with D over S_0. The Berezinian then reduces to a simple expression.

Corollary 1.2. Under the hypotheses just stated, the conditions of 1.1 are equivalent to

$$a_1 + Db_1 = 0. \tag{1.2.1}$$

Proof. Write

$$(2a_1 + Db_1) - a_1 = 0.$$

Of course, in the above, a and a_1 are even, while b and b_1 are odd.

Remark 1.3. (i) For X/S smooth of relative dimension $(1, 1)$, let ℓ be a basis of ω_X and let

$$\delta = D\ell : \mathcal{O} \longrightarrow \omega$$

be a derivation. The condition that D generate a local direct factor of dimension $(1, 0)$ in $T_{X/S}$ is equivalent to the surjectivity of

$$\delta : \Omega^1_{X/S} \longrightarrow \omega_{X/S}.$$

For S an ordinary even scheme, it is also equivalent to the corresponding condition for the morphism induced by δ,

$$\Theta_X^- \longrightarrow \omega_X^-.$$

(ii) For a surjective

$$\delta = D\ell : \Omega^1_{X/S} \longrightarrow \omega_{X/S},$$

if on a schematically dense open set δ is a supercurve structure, then it is one everywhere: in the absence of a reference supercurve structure, the Berezinian in (1.1.1) has an invertible denominator, and since Ber $= 1$ on a schematically dense open set, the identity holds everywhere.

Definition 1.4. A supercurve over S is a superscheme X/S, flat and relatively Cohen-Macaulay, which on a fibrewise dense open set is smooth of relative dimension $(1,1)$, equipped with a derivation

$$\delta : \mathcal{O}_X \longrightarrow \omega_{X/S},$$

where $\omega_{X/S}$ is the dualizing sheaf, such that:

(i) on each geometric fibre, δ induces an isomorphism

$$\mathcal{O}^-_{X_s} \xrightarrow{\sim} \omega^-_{X_s/s};$$

(ii) on the smooth locus, δ is a smooth supercurve structure over S.

By 1.3(ii), condition (ii) may be replaced by:

(ii′) on an open set smooth over S and schematically dense, δ defines a smooth supercurve structure over S.

1.5. For S an ordinary scheme, let us spell out this definition in terms of the scheme

$$C := (X, \mathcal{O}_X^+)$$

over S. The hypotheses of flatness, Cohen-Macaulayness, and fibrewise generically smoothness give

$$C/S \text{ is a flat curve with geometrically reduced fibres.} \tag{1.5.1}$$

The coherent sheaf

$$\mathcal{L} := \mathcal{O}_X^-$$

is flat over S, relatively Cohen-Macaulay, locally free of rank 1 on a fibrewise dense open set, and

$$\mathcal{O}_X^- \cdot \mathcal{O}_X^- = 0.$$

For such a superscheme

$$X = (C, \mathcal{O}_C \oplus \mathcal{L}),$$

the dualizing sheaf ω_X is computed as follows. We have

$$p : X \longrightarrow C,$$

and

$$\omega_{X/S} = p^! \omega_{C/S} = \mathrm{Hom}_{\mathcal{O}_C}(\mathcal{O}_X, \omega_C).$$

The relative Cohen-Macaulay hypothesis ensures the vanishing of the higher Ext's and says that $\omega_{X/S}$ is flat over S and commutes with arbitrary base change. One has

$$\omega^+_{X/S} = \omega_C, \qquad \omega^-_{X/S} = \mathrm{Hom}_{\mathcal{O}_C}(\mathcal{L}, \omega_C).$$

Moreover

$$\mathcal{O}^-_X \omega^+_{X/S} = 0,$$

and

$$\mathcal{O}^-_X \otimes \omega^-_{X/S} \to \omega^+_{X/S}$$

is the evaluation map

$$\mathcal{L} \otimes \mathrm{Hom}(\mathcal{L}, \omega_C) \to \omega_C.$$

In the smooth case, in local coordinates (z, θ), ω_X has basis $[dz/d\theta]$, and ω^-_X is identified with ω_C by

$$\theta[dz/d\theta] \longmapsto dz.$$

A derivation

$$\delta : \mathcal{O}_X \longrightarrow \omega_X$$

is defined by

$$\delta^+ : \mathcal{O}_C \longrightarrow \omega_C \quad \text{(a derivation)},$$

and

$$\delta^- : \mathcal{L} \longrightarrow \mathrm{Hom}(\mathcal{L}, \omega_C) \quad (\mathcal{O}_C\text{-linear}).$$

The derivation conditions give the usual rules when x or y is even. For x and y odd, one should express the symmetry of the bilinear form

$$\mathcal{L} \otimes \mathcal{L} \longrightarrow \omega_C$$

defined by δ^-. Since $\mathcal{L}$ is generically of rank 1, this is automatic. Axiom 1.4(i) says that δ^- is an isomorphism. Axiom 1.4(ii) says that, on the smooth locus,

$$\delta^+ : \mathcal{O}_C \longrightarrow \omega_C$$

is

$$d : \mathcal{O}_C \longrightarrow \Omega^1_C = \omega_C.$$

Indeed, this condition is satisfied by a smooth supercurve in standard local coordinates,

$$\theta\, dz = \theta[dz/d\theta] \sim dz,$$

and all δ satisfying the condition are locally isomorphic for the étale topology. Conversely, in local coordinates, if

$$\delta f = (u(z)\partial_\theta + \theta v(z)\partial_z) f\, [dz/d\theta],$$

then

$$(u(z)\partial_\theta + \theta v(z)\partial_z)^2 = u(z)v(z)\partial_z,$$

and the condition

$$[u(z)v(z)\partial_z/(u(z)\partial_\theta + \theta v(z)\partial_z)] = [\partial_z/\partial_\theta]$$

gives $v = 1$, as desired.

Almost: on a schematically dense open set, one has $\delta^+ = d$; hence this is true everywhere. Thus, over an ordinary scheme, giving a supercurve is equivalent to giving $C, \mathcal{L}$ as in (1.5.1), together with:

$$\text{a bilinear, automatically symmetric, form } \mathcal{L} \otimes \mathcal{L} \longrightarrow \omega_C \tag{1.5.2}$$

which induces an isomorphism

$$\mathcal{L} \xrightarrow{\sim} \mathrm{Hom}(\mathcal{L}, \omega_C).$$

As explained, one then has

$$X = (C, \mathcal{O}_C \oplus \mathcal{L}), \qquad \omega_X = \omega_C \oplus \mathcal{L}, \tag{1.5.3}$$

with the module structure defined by (1.5.2), and

$$\delta = (d, \mathrm{id}).$$

In characteristic 2, a family of smooth supercurves is not necessarily locally, even formally, trivial. For this reason I shall henceforth assume that 2 is invertible.

So that everything up to this point could work over **Z**, I used the definition of a “commutative super-ring” in which $a^2 = 0$ for odd a, and I took care to use D^2 rather than $[D, D]$ for odd derivations.

1.6. The following description of $\omega_{X/S}$ will be useful. Let

$$j : U \hookrightarrow X$$

be a fibrewise dense smooth open subset. For every section s of $j_*\omega_{U/S}$ and every connected component F of $X - U$, finite over S, a residue

$$\mathrm{Res}_F(s) \in \mathcal{O}_S$$

is defined. The sheaf $\omega_{X/S}$ is the subsheaf of $j_*\omega_{U/S}$ consisting of those sections s such that, locally for the étale topology on S and for every such F, and every f in $\mathcal{O}_X$, one has

$$\mathrm{Res}_F(fs) = 0.$$

Over nilpotent S, Res_F is a sum of residues along the branches passing through F and, for each branch $B \hookrightarrow X$ closed in X, is invariant under birational morphisms $B_1 \to B$. This permits one to compute ω_X in Serre’s manner.

Proposition 1.7. Let X_0 be a supercurve over a field k. An infinitesimal deformation X of X_0 that is trivial as a deformation of the underlying superscheme is trivial as a deformation of supercurve.

Proof. It is enough to treat first-order deformations, by a Schlessinger-type argument. The case reduces to an even deformation and an odd deformation.

a. Even deformation. We use the point of view of 1.5. Suppose that X_0 is defined by C, $\mathcal{L}$, and an isomorphism

$$\mathcal{L} \xrightarrow{\sim} \mathrm{Hom}(\mathcal{L}, \omega_C).$$

A deformation with the superscheme held fixed can be viewed as a deformation of this isomorphism. Such a deformation is of the form $\varphi_0 \circ a$, where a is an infinitesimal automorphism of $\mathcal{L}$. Such an infinitesimal automorphism $a = 1 + a_1$ has square root $1 + \frac{1}{2}a_1$, and φ is conjugate to φ_0 by $(1+a_1)^{-1/2}$. It suffices to check this on the smooth locus, where a is multiplication by an element of $\mathcal{O}_{C_0[t]/(t^2)}$ congruent to 1 modulo t. Conjugating φ does not change the isomorphism class of the deformation of supercurve, so the deformation is trivial.

b. Odd deformations. A deformation of (X_0, δ_0), trivial as a deformation of superscheme, is of the form (X, δ), with X obtained from X_0 by extension of scalars and

$$\delta = \delta_0 + \varepsilon D,$$

where $\delta_0 : \Omega^1_X \to \omega_X$ is obtained from the original δ_0 by extension of scalars, and where D is an odd derivation

$$\mathcal{O}_{X_0} \longrightarrow \omega_{X_0}.$$

We must prove that, if δ satisfies 1.4(ii), then δ is conjugate to δ_0 by an automorphism of X trivial on X_0. Such an automorphism is of the form

$$f \longmapsto f + \varepsilon E f,$$

where E is an odd derivation of $\mathcal{O}_{X_0}$ into $\mathcal{O}_{X_0}$.

From the point of view of 1.5, an odd derivation

$$D : \mathcal{O}_{X_0} \longrightarrow \omega_{X_0}$$

is given by its components

$$\begin{cases} D_0 : \mathcal{O}_C \longrightarrow \mathcal{L}, & \text{a derivation,} \\ D_1 : \mathcal{L} \longrightarrow \omega_C, & D_1(a\ell) = -D_0(a)\ell + aD_1(\ell), \end{cases}$$

where the underlined term comes from the product $\mathcal{L} \otimes \mathcal{L} \to \omega_C$.

For $\delta_0 + \varepsilon D$ satisfying 1.4(ii), D_1 is determined by D_0. It is enough to verify this on the smooth locus. There, in local coordinates,

$$Df = (u(z)\partial_z + v(z)\theta\partial_\theta) f \,[dz/d\theta],$$

and

$$D_0 : f \longmapsto u(z)\partial_z f \,[dz/d\theta] : \Theta^+_{X_0} \to \omega^-_{X_0}.$$

Writing

$$D = \theta v(\partial_\theta + \theta\partial_z) + u\partial_z,$$

and applying 1.2, one sees that 1.4(ii) means

$$v = \partial_z u.$$

Thus D_0 determines D.

An odd derivation E of $\mathcal{O}_X$ is described, from the point of view of 1.5, by its components

$$\begin{cases} E_0 : \mathcal{O}_C \longrightarrow \mathcal{L}, & \text{a derivation,} \\ E_1 : \mathcal{L} \longrightarrow \mathcal{O}_C, & \mathcal{O}_C\text{-linear.} \end{cases}$$

We take for E the derivation with components $(D_0, 0)$. To verify that the infinitesimal automorphism id $+\varepsilon E$ defined by E conjugates δ_0 to δ, it is enough to check this on the smooth locus, and it is enough to verify that the component 0 of $L_E(\delta_0)$ is D_0. In local coordinates one has

$$E = u(z)\theta\partial_z, \qquad \delta_0 = (\partial_\theta + \theta\partial_z)[dz/d\theta].$$

Thus

$$L_E(\delta_0) = [E, \partial_\theta + \theta\partial_z][dz/d\theta] + (\partial_\theta + \theta\partial_z)L_E([dz/d\theta]),$$

and

$$[E, \partial_\theta + \theta\partial_z] = [u(z)\theta\partial_z, \partial_\theta + \theta\partial_z] = u(z)\partial_z.$$

The second term contributes to component 1 of $L_E(\delta_0)$. We have recovered the required D_0.

~~1.8. Nevertheless. I have finally understood what condition (ii) of Definition 1.4 means: that δ is its own transpose (up to a sign to be added).~~